

Enzyme Action: Preparing the pH Investigation

Today I can... ☐ Use enzyme, substrate, active site and specificity accurately in an explanation ☐ Explain how an unsuitable pH can change an active site's shape and reduce enzyme activity ☐ Identify pH as the independent variable, time to starch disappearance as the dependent variable, and suitable control variables
Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____



LAUNCH · GCSE Biology · Enzyme Action: Preparing the pH Investigation

LAUNCH SCIENCE | last lesson |

Enzyme action and pH

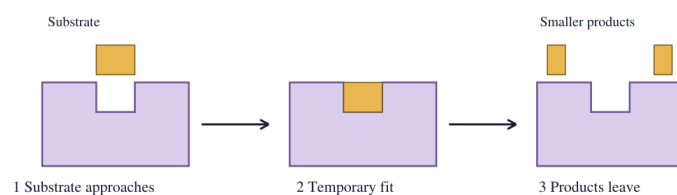
Knowledge organiser

Learning objective

I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

Success criteria

1. Use enzyme, substrate, active site and specificity accurately in an explanation.
2. Explain how an unsuitable pH can change an active site's shape and reduce enzyme activity.
3. Identify pH as the independent variable, time to starch disappearance as the dependent variable, and suitable control variables.



Active site: a specific region where substrate binds. Enzyme remains available.

A complementary active site

Specificity means acting on particular substrates. Amylase catalyses starch breakdown into smaller sugars. Products leave; the enzyme remains available. This simplified shape model does not show literal rigid molecules.

Connect pH to a measured endpoint

pH describes acidity or alkalinity. Unsuitable pH can alter the active site and reduce activity. Denaturation disrupts the enzyme's three-dimensional structure and function. Slower activity alone does not prove permanent denaturation.

Plan a valid comparison

Change pH. Time the first sampled drop in which iodine remains orange-brown (starch not detected). Blue-black means starch detected. Match temperature, concentrations, volumes and sampling. check separate drops; keep iodine out of the main mixture.

Read the evidence honestly

A no-active-amylase comparison should remain starch-positive. Record "no endpoint by 300 s" when that is all that was observed; do not invent zero time or zero rate. The optimum pH is to be investigated.

Made by Matt | Use the reference to explain, calculate and evaluate.